

PAST EXAMINATION QUESTIONS (FP1)

TOPIC 6: PROOF BY INDUCTION

1. It is given that $y = \ln(ax + 1)$, where a is a positive constant. Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^n y}{dx^n} = (-1)^{n-1} \frac{(n-1)! a^n}{(ax + 1)^n}. \quad [6]$$

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2. (i) Prove by mathematical induction that, for $z \neq 1$ and all positive integers n ,

$$1 + z + z^2 + \dots + z^{n-1} = \frac{z^n - 1}{z - 1}. \quad [5]$$

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- (ii) By letting $z = \frac{1}{2}(\cos \theta + i \sin \theta)$, use the de Moivre's theorem to deduce that

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m \sin m\theta = \frac{2 \sin \theta}{5 - 4 \cos \theta}. \quad [5]$$

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3. Prove by mathematical induction that $3^{3n} - 1$ is divisible by 13 for every positive integer n . [5]

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4. The sequence of positive numbers $u_1, u_2, u_3, \dots$ is such that $u_1 < 3$ and, for $n \geq 1$,

$$u_{n+1} = \frac{4u_n + 9}{u_n + 4},$$

- (i) By considering $3 - u_{n+1}$, or otherwise, prove by mathematical induction that $u_n < 3$ for all positive integers n . [5]
 (ii) Show that $u_{n+1} > u_n$ for $n \geq 1$. [3]

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5. It is given that $y = e^x u$, where u is a function of x . The r th derivatives $\frac{d^r y}{dx^r}$ and $\frac{d^r u}{dx^r}$ are denoted by $y^{(r)}$ and $u^{(r)}$ respectively. Prove by mathematical induction that, for all positive integers n ,

$$y^{(n)} = e^x \left(\binom{n}{0} u + \binom{n}{1} u^{(1)} + \binom{n}{2} u^{(2)} + \dots + \binom{n}{r} u^{(r)} + \dots + \binom{n}{n} u^{(n)} \right). \quad [8]$$

[You may use without proof the result that $\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}$.]

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6. It is given that $f(n) = 2^{3n} + 8^{n-1}$. By simplifying $f(k) + f(k+1)$, or otherwise, prove by mathematical induction that $f(n)$ is divisible by 9 for every positive integer n . [6]

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7. For the sequence $u_1, u_2, u_3, \dots$, it is given that $u_1 = 8$ and

$$u_{r+1} = \frac{5u_r - 3}{4}$$

for all r .

- (i) Prove by mathematical induction that

$$u_n = 4\left(\frac{5}{4}\right)^n + 3,$$

for all positive integers n .

[5]

- (ii) Deduce the set of values of x for which the infinite series

$$(u_1 - 3)x + (u_2 - 3)x^2 + \dots + (u_r - 3)x^r + \dots$$

is convergent. ($|x| < 4.5$)

[2]

- (iii) Use the result given in part (i) to find surds a and b such that

$$\sum_{n=1}^N \ln(u_n - 3) = N^2 \ln a + N \ln b.$$

$$\left(a = \frac{\sqrt{5}}{2}, b = 2\sqrt{5}\right)$$

[3]

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8. (i) Show that $\frac{d^{n+1}}{dx^{n+1}}(x^{n+1} \ln x) = \frac{d^n}{dx^n}(x^n + (n+1)x^n \ln x)$.

[2]

- (ii) Prove by mathematical induction that, for all positive integers n ,

$$\frac{d^n}{dx^n}(x^n \ln x) = n! \left(\ln x + 1 + \frac{1}{2} + \dots + \frac{1}{n} \right).$$

[5]

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9. Prove, by mathematical induction that, that $5^n + 3$ is divisible by 4 for all non-negative integers n .

[5]

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10. Prove, by mathematical induction, that

$$\sum_{r=1}^n r \ln \left(\frac{r+1}{r} \right) = \ln \left(\frac{(n+1)^n}{n!} \right)$$

for all positive integers n .

[6]

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11. Using factorials, show that $\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}$.

[2]

Hence prove by mathematical induction that

$$(a+x)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}x + \dots + \binom{n}{r}a^{n-r}x^r + \dots + \binom{n}{n}x^n$$

for every positive integer n .

[4]

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12. Prove by mathematical induction that, for all positive integers n , $10^n + 3 \times 4^{n+2} + 5$ is divisible by 9. [6]

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13. It is given that a diagonal of a polygon is a line joining two non-adjacent vertices. Prove, by mathematical induction, that an n -sided polygon has $\frac{1}{2}n(n-3)$ diagonals, where $n \geq 3$. [6]

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14. Given that a is a constant, prove by mathematical induction that, for every positive integer n ,

$$\frac{d^n}{dx^n}(xe^{ax}) = na^{n-1}e^{ax} + a^nxe^{ax}. \quad [6]$$

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15. The sequence $a_1, a_2, a_3, \dots$ is such that $a_1 > 5$ and $a_{n+1} = \frac{4a_n}{5} + \frac{5}{a_n}$ for every positive integer n . Prove by mathematical induction that $a_n > 5$ for every positive integer n . [5]

Prove also that $a_n > a_{n+1}$ for every positive integer n .

[2]

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16. Prove that mathematical induction that, for all positive integers n ,

$$\sum_{r=1}^n \frac{1}{(2r)^2 - 1} = \frac{n}{2n+1}. \quad [6]$$

State the value of

$$\sum_{r=1}^{\infty} \frac{1}{(2r)^2 - 1}. \quad [1]$$

$\left(\frac{1}{2}\right)$

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17. It is given that $u_r = r \times r!$ for $r = 1, 2, 3, \dots$. Let $S_n = u_1 + u_2 + u_3 + \dots + u_n$. Write down the values of

$$2! - S_1, \quad 3! - S_2, \quad 4! - S_3, \quad 5! - S_4. \quad [2]$$

Conjecture a formula for S_n . ($S_n = (n+1)! - 1$) [1]

Prove, by mathematical induction, a formula for S_n , for all positive integers n . [4]

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18. Prove by mathematical induction that, for all non-negative integers n ,

$$11^{2n} + 25^n + 22$$

is divisible by 24. [6]

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19. It is given that $\phi(n) = 5^n(4n + 1) - 1$, for $n = 1, 2, 3, \dots$. Prove, by mathematical induction, that $\phi(n)$ is divisible by 8, for every positive integer n . [7]

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20. It is given that $y = (1 + x)^2 \ln(1 + x)$. Find $\frac{d^3 y}{dx^3}$. [3]

Prove by mathematical induction that, for every integer $n \geq 3$,

$$\frac{d^n y}{dx^n} = (-1)^{n-1} \frac{2(n-3)!}{(1+x)^{n-2}}. \quad [5]$$

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21. *****Remove from 9231/1 AS syllabus*****

Prove by mathematical induction that, for every positive integer n ,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. \quad [5]$$

Express $\sin^5 \theta$ in the form $p \sin 5\theta + q \sin 3\theta + r \sin \theta$, where p, q and r are rational numbers to be determined. $\left(p = \frac{1}{16}, q = -\frac{5}{16}, r = \frac{10}{16}\right)$ [6]

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22. Prove by mathematical induction that $5^{2n} - 1$ is divisible by 8 for every positive integer n . [5]

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23. Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^n}{dx^n} (e^x \sin x) = (\sqrt{2})^n e^x \sin \left(x + \frac{1}{4} n\pi\right). \quad [7]$$

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24. *****Remove from 9231/1 AS syllabus*****

Let I_n denote $\int_0^\infty x^n e^{-2x} dx$. Show that $I_n = \frac{1}{2} n I_{n-1}$, for $n \geq 1$. [2]

Prove by mathematical induction that, for all positive integers n , $I_n = \frac{n!}{2^{n+1}}$. [6]

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25. Let $S_n = \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{N}{(N+1)!}$. Prove by mathematical induction that, for all positive integers N ,

$$S_N = 1 - \frac{1}{(N+1)!}. \quad [5]$$

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25. Prove, by mathematical induction, that, for integers $n \geq 2$,

$$4^n > 2^n + 3^n. \quad [5]$$

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26. For the sequence $u_1, u_2, u_3, \dots$. It is given that $u_1 = 1$ and $u_{r+1} = \frac{3u_r - 2}{4}$ for all r . Prove by mathematical induction that $u_n = 4\left(\frac{3}{4}\right)^n - 2$, for all positive integers n . [5]

May/June 2012/13/2

27. Prove by mathematical induction that, for all positive integers n ,

$$\frac{d^n}{dx^n}(e^x \sin x) = 2^{\frac{1}{2}\pi} e^x \sin\left(x + \frac{1}{4}n\pi\right). \quad [7]$$

Oct/Nov 2011/11/12/3

28. Prove by mathematical induction that, for all positive integers n ,

$$\frac{d^n}{dx^n}\left(\frac{1}{2x+3}\right) = (-1)^n \frac{n! 2^n}{(2x+3)^{n+1}}. \quad [6]$$

Oct/Nov 2011/13/2

29. It is given that $f(n) = 3^{3n} + 6^{n-1}$.

(i) Show that $f(n+1) + f(n) = 28(3^{3n}) + 7(6^{n-1})$. [2]

(ii) Hence, or otherwise, prove by mathematical induction that $f(n)$ is divisible by 7 for every positive integer n . [4]

May/June 2011/11/12/4

30. Let $A = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$. Prove by mathematical induction that, for every positive integer n ,

$$A^n = \begin{pmatrix} 2^n & 3(2^n - 1) \\ 0 & 1 \end{pmatrix} \quad [5]$$

May/June 2011/13/2

31. Prove by mathematical induction that, for all non-negative integers n , $7^{2n+1} + 5^{n+3}$ is divisible by 44. [5]

Oct/Nov 2010/1/4

32. *****Remove from 9231/1 AS syllabus*****

Let

$$I_n = \int_1^e x(\ln x)^n dx, \quad [3]$$

where $n \geq 1$. Show that

$$I_{n+1} = \frac{1}{2}e^2 - \frac{1}{2}(n+1)I_n.$$

Hence prove by induction that, for all positive integers n , I_n is of the form $A_n e^2 + B_n$, where A_n and B_n are rational numbers. [6]

May/June 2010/11/12/5

33. The sequence $x_1, x_2, x_3, \dots$ is such that $x_1 = 3$ and

$$x_{n+1} = \frac{2x_n^2 + 4x_n - 2}{2x_n + 3}$$

for $n = 1, 2, 3, \dots$. Prove by induction that $x_n > 2$ for all n .

[6]

May/June 2010/13/3

34. Prove by induction that

$$\sum_{n=1}^N n^3 = \frac{1}{4} N^2 (N+1)^2. \quad [5]$$

Use this result, together with the formula for $\sum_{n=1}^N n^2$, to show that

$$\sum_{n=1}^N (20n^3 + 36n^2) = N(N+1)(N+3)(5N+2). \quad [3]$$

Let

$$S_N = \sum_{n=1}^N (20n^3 + 36n^2 + \mu n).$$

Find the value of the constant μ such that S_N is of the form $N^2(N+1)(aN+b)$, where the constants a and b are to be determined. ($\mu = -12, a = 5, b = 17$)

[3]

Show that, for this value of μ ,

$$5 + \frac{22}{N} < N^{-4} S_N < 5 + \frac{23}{N},$$

for all $N \geq 18$.

[3]

Oct/Nov 2009/1/11E

35. ***Remove from 9231/1 AS syllabus***

Let

$$I_n = \int_0^1 t^n e^{-t} dt,$$

where $n \geq 0$. Show that, for all $n \geq 1$,

$$I_n = nI_{n-1} - e^{-1}. \quad [3]$$

Hence prove by induction that, for all positive integers n ,

$$I_n < n!. \quad [5]$$

May/June 2009/1/7

36. Use the induction to prove that

$$\sum_{n=1}^N \frac{4n+1}{n(n+1)(2n-1)(2n+1)} = 1 - \frac{1}{(N+1)(2N+1)}. \quad [6]$$

Show that,

$$\sum_{n=N+1}^{2N} \frac{4n+1}{n(n+1)(2n-1)(2n+1)} < \frac{3}{8N^2}. \quad [4]$$

Oct/ Nov 2008/1/9

37. Prove by induction that

$$\sum_{r=1}^n (3r^5 + r^3) = \frac{1}{2}n^3(n+1)^3,$$

for all $n \geq 1$. [5]

Use this result together with the List of Formulae (MF10) to prove that

$$\sum_{r=1}^n r^5 = \frac{1}{12}n^2(n+1)^2Q(n),$$

where $Q(n)$ is a quadratic function of n which is to be determined. ($Q(n) = 2n^2 + 2n - 1$) [3]
May/June 2008/1/7

38. Prove by induction that, for all $n \geq 1$,

$$\frac{d^n}{dx^n}(e^{x^2}) = P_n(x)e^{x^2},$$

where $P_n(x)$ is a polynomial in x of degree n with the coefficient of x^n equal to 2^n . [6]
Oct/Nov 2007/1/3

39. The sequence $x_1, x_2, x_3, \dots$ is such that $x_1 = 1$ and

$$x_{n+1} = \frac{1 + 4x_n}{5 + 2x_n}.$$

Prove by induction that $x_n > \frac{1}{2}$ for all $n \geq 1$. [5]

Prove also that $x_n > x_{n+1}$ for all $n \geq 1$. [3]

May/June 2007/1/8

40. Prove by mathematical induction that, for all positive integers n , $10^{3n} + 13^{n+1}$ is divisible by 7. [5]

Oct/Nov 2016/1/4

41. Prove by induction, or otherwise, that

$$23^{2n} + 31^{2n} + 46$$

is divisible by 48, for all integers $n \geq 0$.

[6]

May/June 2006/1/3

42. The sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = 1$ and

$$u_{n+1} = -1 + \sqrt{u_n + 7}.$$

- (i) Prove by induction that $u_n < 2$ for all $n \geq 1$. [4]
 (ii) Show that if $u_n = 2 - \varepsilon$, where ε is small, then

$$u_{n+1} \approx 2 - \frac{1}{6}\varepsilon. \quad [2]$$

Oct/Nov 2005/1/2

43. ***Remove from 9231/1 AS syllabus***

The integral I_n , where n is a non-negative integer, is defined by

$$I_n = \int_0^1 e^{-x}(1-x)^n dx.$$

- (i) Show that $I_{n+1} = 1 - (n+1)I_n$. [3]
 (ii) Use induction to show that I_n is one of the form $A_n + B_n e^{-1}$, where A_n and B_n are integers. [4]
 (iii) Express B_n in terms of n . [2]

May/June 2005/1/8

44. The sequence of real numbers $a_1, a_2, a_3, \dots$ is such that $a_1 = 1$ and

$$a_{n+1} = \left(a_n + \frac{1}{a_n}\right)^\lambda,$$

where λ is a constant greater than 1. Prove by mathematical induction that, for $n \geq 2$,

$$a_n \geq 2^{g(n)},$$

where $g(n) = \lambda^{n-1}$. [6]

Prove also that, for $n \geq 2$, $\frac{a_{n+1}}{a_n} > 2^{(\lambda-1)g(n)}$. [3]

Oct/Nov 2004/1/8

45. It is given that

$$\frac{d^n}{dx^n} \left(\frac{\ln x}{x} \right) = \frac{a_n \ln x + b_n}{x^{n+1}},$$

where a_n and b_n depend only on n .

(i) Find a_1, a_2 and a_3 . ($-1, 2, -6$)

[3]

(ii) Use mathematical induction to establish a formula for a_n . ($a_n = (-1)^n n!$)

[5]

May/June 2004/1/4

46. Given that $y = x \sin x$, find $\frac{d^2 y}{dx^2}$ and $\frac{d^4 y}{dx^4}$, simplifying your results as far as possible, and show that

$$\frac{d^6 y}{dx^6} = -x \sin x + 6 \cos x. \quad [3]$$

Use induction to establish an expression for $\frac{d^{2n} y}{dx^{2n}}$, where n is a positive integer. [5]

Oct/Nov 2003/1/4

47. Prove by induction that, for all $N \geq 1$,

$$\sum_{n=1}^N \frac{n+2}{n(n+1)^{2n}} = 1 - \frac{1}{(N+1)2^N}. \quad [5]$$

May/June 2003/1/2

48. It is given that, for $n = 0, 1, 2, 3, \dots$

$$a_n = 17^{2n} + 3(9)^n + 20.$$

Simplify $a_{n+1} - a_n$, and hence prove by induction that a_n is divisible by 24 for all $n \geq 0$.
($288(17^{2n}) + 24(9^n)$)

[6]

Oct/Nov 2002/1/3

49. The sequence of positive numbers $u_1, u_2, u_3, \dots$ is such that $u_1 < 4$ and

$$u_{n+1} = \frac{5u_n + 4}{u_n + 2}.$$

By considering $4 - u_{n+1}$, or otherwise, prove by induction that $u_n < 4$ for $n \geq 1$. [5]

Prove also that $u_{n+1} > u_n$ for all $n \geq 1$.

[3]

May/June 2002/1/6